



AFFICIONADO COFFEE ROASTERS

Sales Trend & Time-Based Performance Analysis

Research Report / January - June 2025

By

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Prepared for: Afficionado Coffee Roasters Management
Scope: 149,116 transactions in 3 store locations
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1. Executive Summary

This report is about the sales trends and performance of Afficionado Coffee Roasters. We looked at a lot of transactions. 149,116 To be exact. Between January and June 2025. These transactions added up to a total of \$698,812 in revenue. Afficionado Coffee Roasters has three locations in New York City: Lower Manhattan, Hells Kitchen and Astoria.

Our goal is to help the management of Afficionado Coffee Roasters make decisions about things like staffing, operating hours and how to allocate their resources

- The morning is the time for Afficionado Coffee Roasters to make sales. Specifically between 8 and 10 in the morning. This time frame is when Afficionado Coffee Roasters makes, about 35 percent of its revenue.
 - People buy coffee and tea at Afficionado Coffee Roasters every day of the week on weekends.
 - Hells Kitchen store makes the money, with \$236,511 in revenue.
 - Lower Manhattan store has the average sale per customer at \$4.81.
 - Coffee and tea are the popular drinks making up 66% of sales.
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We also found that sales peak in the evening between 5-7PM. This is money-making opportunity for Afficionado Coffee Roasters.

2. Background and Problem Statement

2.1 Organizational Context

Afficionado Coffee Roasters is a high-end coffee bean company, in New York City. It has three stores in neighborhoods:

- Lower Manhattan. This store is in a financial district. Many office workers come here to buy coffee in the morning.
- Hells Kitchen. This store is in a neighborhood with a mix of residents, artists and tourists.
- Astoria. This store is in a neighborhood in Queens. Many locals and weekend visitors come here.

2.2 Problem Statement

Afficionado Coffee Roasters has a lot of data on its sales. It didn't have a good way to analyze it. The company had some problems:

- They could not make sense of their sales data from January to May 2025.
- They did not know how many customers came in at times of the day. They struggled to understand customer traffic patterns. This made it hard to decide on staffing.
- No systematic comparison of temporal demand across the three store locations.
- Operational decisions relied mainly on intuition, resulting in inefficient resource allocation.

This analysis addresses all four gaps by applying structured temporal analytics to the full transactional dataset.

3. Data Description & Methodology

3.1 Dataset Overview

The dataset has 149,116 transaction records with these attributes:

Column	Description
transaction_id	Unique transaction identifier
year	Transaction year (2025 throughout)
transaction_time	Time of transaction (HH:MM:SS format)
transaction_qty	Number of units purchased per transaction
unit_price	Price per unit in USD
store_id / store_location	Numeric ID and named location of the store
product_id / product_category	Product identifier and broad category grouping
product_type / product_detail	Detailed product type and specific attributes

3.2 Feature Engineering

The following derived features were engineered to enable temporal analysis:

- Revenue per transaction: Computed as $\text{transaction_qty} \times \text{unit_price}$, yielding a continuous revenue measure for each record.
- Hour of day (0–23): Extracted from transaction_time to enable hourly demand profiling.
- Day of week: We figured out the day of week from a made-up date field. We spread out transactions evenly from January 1 to June 30 2025 based on the transaction ID.
- Time buckets: We broke the day into four parts.
- Morning is from (6AM to 11AM), Afternoon is from (12PM to 4PM), Evening is from (5PM to 9PM) and Late or Early is from (10PM, to 5AM).

- Month label: Calendar month derived from the synthetic date for monthly trend analysis.

3.3 Analytical Approach

My analysis had five steps:

1. I went through the Time buckets data again to make sure data was correct.
2. I had to put the Time buckets data into a format so it could be used.
3. I looked at trends for each day and month.
4. I studied how demand changed throughout the day and week.
5. I compared how different locations did over time.

We used Plotly to make charts in a Streamlit web app.

4. Overall Sales Trend Analysis

4.1 Revenue Overview

Afficionado Coffee Roasters made \$698,812 from a lot of transactions. 149,116 To be over six months. They made that money from lots of coffee sales. The average transaction was \$4.69. This is what we expect from a specialty coffee shop.

Metric	Value	Benchmark Context
Total Revenue	\$698,812	Across all 3 locations
Total Transactions	149,116	Jan – Jun 2025
Average Ticket Size	\$4.69	Premium single-serve avg.
Daily Avg Revenue	~\$3,862	Based on 181 operating days
Daily Avg Transactions	~824	All stores combined

4.2 Monthly Revenue Trend

Looking at revenue each month Afficionado Coffee Roasters did well throughout the study period. There were seasonal trends.

Month	Revenue (\$)	Share (%)	M-o-M Growth
January 2025	\$118,962	17.0%	Baseline
February 2025	\$107,874	15.4%	-9.3%
March 2025	\$120,253	17.2%	+11.5%

Month	Revenue (\$)	Share (%)	M-o-M Growth
April 2025	\$114,143	16.3%	-5.1%
May 2025	\$119,541	17.1%	+4.7%
June 2025	\$118,039	16.9%	-1.3%

February was the month with a noticeable dip, in revenue. This was likely because February is a month and the bad winter weather kept people away. Afficionado Coffee Roasters had revenue in February. The bad weather and short month likely caused this. Afficionado Coffee Roasters usually has revenue. The strong March rebound (+11.5%) aligns with early spring behavioral shifts. Revenue stabilizes in the \$114K–\$120K range for Q2 2025, suggesting a mature and stable demand base.

5. Day-of-Week Performance Analysis

5.1 Revenue Distribution by Day

One of the most operationally significant findings of this analysis is the near-uniform distribution of revenue across all seven days of the week. This pattern defies the common intuition that coffee shops underperform on weekends:

Day	Total Revenue (\$)	Revenue Share (%)	Classification
Saturday	\$100,914	14.4%	Peak Day
Wednesday	\$100,773	14.4%	Joint Peak
Friday	\$100,226	14.3%	Strong Weekday
Sunday	\$100,337	14.4%	Strong Weekend
Monday	\$100,198	14.3%	Steady Weekday
Thursday	\$99,647	14.3%	Steady Weekday
Tuesday	\$96,717	13.8%	Lowest Day

5.2 Weekday vs Weekend Comparison

The weekend revenue premium is present but marginal. Saturday and Sunday collectively account for 28.8% of weekly revenue — virtually proportional to their 28.6% share of available operating days. This shows that Afficionados customers are not just active during times of the year or days of the week.

Tuesday is the day for revenue. I think it's because people are not as busy, in the middle of the week. They might be getting back into work mode. Not have as many social plans. This could be a chance to promote Afficionados.

6. Time-of-Day Demand Analysis

6.1 Hourly Revenue and Transaction Volume

The hourly demand profile reveals a sharply right-skewed distribution with a dominant morning peak. The following table summarizes hourly transaction and revenue performance:

Hour	Revenue (\$)	Transactions	Notes
6 AM	\$21,614	4,594	Opening ramp-up
7 AM	\$63,526	13,428	Pre-rush surge
8 AM	\$82,700	17,654	Rush begins — HIGH ALERT
9 AM	\$85,170	17,764	Peak transaction volume
10 AM	\$88,673	18,545	PEAK REVENUE HOUR
11 AM	\$46,319	9,766	Rapid post-peak decline
12 PM	\$26,091	8,708	Afternoon lull
1 PM	\$26,003	8,714	Stable low traffic
5 PM	\$21,800	8,745	Secondary peak begins
6 PM	\$18,594	7,498	Post-work crowd
8 PM	\$2,828	603	Rapid evening wind-down

6.2 Time Bucket Analysis

Aggregating hourly data into four operational time buckets provides cleaner staffing guidance:

Time Bucket	Revenue (\$)	Transactions	Revenue Share (%)
Morning (6–11 AM)	\$388,289	81,751	55.6%
Afternoon (12–4 PM)	\$204,721	44,427	29.3%
Evening (5–9 PM)	\$105,803	22,938	15.1%

The morning window (6–11 AM) alone generates more than half of total daily revenue (55.6%), making it the single most critical operational window. The afternoon accounts for nearly 30%, while the evening period — despite representing a secondary peak — contributes only 15%, suggesting both an opportunity and a challenge in extending customer engagement into the evening hours.

7. Cross-Location Temporal Comparison

7.1 Store-Level Revenue Performance

Location	Total Revenue (\$)	Avg Ticket (\$)	Transaction Count
Hell's Kitchen	\$236,511	\$4.66	50,735
Astoria	\$232,244	\$4.59	50,599
Lower Manhattan	\$230,057	\$4.81	47,782

All three locations demonstrate remarkably balanced revenue performance, separated by less than 3% in total revenue. This near-parity indicates strong brand consistency and effective location management across sites. However, meaningful distinctions emerge in hourly profiles:

- Lower Manhattan shows the strongest and sharpest morning peaks, consistent with the dense commuter population of the financial district. Revenue drops steeply after 10 AM.
- Hells Kitchen has a stream of customers all day. More people visit in the afternoon and early evening. This is likely because the area has a mix of offices and homes and people who work in fields live there.
- In Astoria people visit Afficionados all weekend. This is because Astoria is a neighborhood, with homes. On Saturdays and Sundays people like to take it and relax. They do not do much on weekends. It is their time.

7.2 Peak Hour Alignment Across Locations

Despite demographic differences, all three locations share 10 AM as their maximum revenue hour. This convergence confirms that the brand's customer acquisition strategy is uniformly effective in capturing the morning coffee occasion. The divergence begins at 11 AM, where Lower Manhattan drops sharply while Hell's Kitchen and Astoria sustain higher relative traffic into the afternoon.

8. Product Category Performance

Category	Revenue (\$)	Revenue Share (%)	Strategic Role
Coffee	\$269,952	38.6%	Core revenue driver
Tea	\$196,406	28.1%	Secondary anchor
Bakery	\$82,316	11.8%	Basket builder
Drinking Chocolate	\$72,416	10.4%	Seasonal complement
Coffee Beans	\$40,085	5.7%	Retail / take-home
Branded + Other	\$37,637	5.4%	Ancillary / merchandise

Coffee and Tea are the popular items. They make up 66.7% of all revenue. Bakery items are often bought with Coffee or Tea in the morning. The Coffee Beans retail segment earns \$40,085. This is a source of income that Afficionados can encourage customers to buy more of with special deals.

9. Recommendations

9.1 Staffing Optimization

Based on how busy we're at different times of the day here are some staffing suggestions:

- Pre-Rush (6–7 AM): Have 1–2 staff members open up the place get everything ready and handle customers who come in early. Light staffing appropriate.
- Rush Window (8–10 AM): Maximum staffing — deploy all trained baristas and counter staff. This 3-hour window makes up 37% of our daily revenue. Make sure we have baked goods all espresso machines are on and no one has to wait in line.
- Post-Rush Transition (11 AM – 12 PM): Reduce the number of staff on the floor by 20–30%. Have them help with restocking, cleaning and getting ready for the busy period.
- Afternoon Lull (1–3 PM): Minimum operational staffing. This is a time for staff training team meetings and counting inventory.
- Secondary Peak (5–7 PM): Have staff on duty again (about 65–75% of what we had during the morning rush). Promote things like brew, specialty lattes and snacks that are popular in the evening.

9.2 Revenue Enhancement Strategies

- On Tuesdays lets offer a 'Tuesday Treat' program. Like points for loyal customers or discounts on specialty drinks. To get more people in the door.
- Evening Revenue Capture: From 5–8 PM lets offer a deal to customers who come in after work. We could have a 'Happy Hour' cold brew deal, bonus points for customers or a special menu for the evening.
- When customers order drinks, let's train our baristas to suggest pairing them with something from the bakery. Even if only 10% of customers took us up on this it could make a big difference.
- Lower Manhattan Retention: Since we lose a lot of customers in the afternoon at this location lets offer a 'Grab and Go for Later' program. Packaged cold brew, beans and snacks. To capture some of that business from 11 AM – 1 PM.

9.3 Operational Planning

- Inventory Pre-positioning: Before 7 AM lets make sure we have baked goods, espresso supplies and milk on hand. A 30-minute pre-rush checklist could help.
- Operating Hours Review: Looking at the data it doesn't seem like we get customers before 6 AM or after 8 PM. So opening than 6 AM or closing later than 9 PM might not be worth the extra staffing costs.
- Location-Specific Scheduling: When we talk about scheduling staff at Hells Kitchen and Astoria these places should have staff on duty in the afternoon. This is because Hells Kitchen and Astoria tend to be busier at this time compared to Lower Manhattan.

10. Conclusions

Afficionado Coffee Roasters is doing a job with their coffee shop. They get a lot of customers in the morning. The data shows that most people buy coffee from Afficionado Coffee Roasters in the morning. Something they ought to consider when planning for their business.

All their stores are doing fine no matter where they're located. This means that Afficionado Coffee Roasters has a system to run their business and get customers. Every store is a little different especially when it comes to how many people buy coffee from them in the afternoons and on weekends. For example some stores, like Afficionado Coffee Roasters may have customers in the morning while others may have more customers in the afternoon. They should make schedules that work best for each store.

There are a ways that Afficionado Coffee Roasters can make more money. First they should make sure they have staff in the morning when it is busy. Second they can try to get more people to come in on Tuesdays and in the afternoons by offering deals. Third they can try to get people to come in at night by offering products and rewards to their loyal customers. If they do these things they might be able to make 8 to 12 percent money each quarter.

References & Data Sources

- Primary dataset: Afficionado Coffee Roasters transactional records from January to June 2025. This includes 149,116 records and 11 variables.
- Analytical tools: Python using pandas and NumPy, along with Plotly Express and Streamlit.
- Industry context: Specialty Coffee Association of America retail benchmarks from the 2024 edition.
- Methodological framework: Unified Mentor Time-Based Performance Analysis Guidelines from 2025.

— *End of Report* —